```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()

filename = next(iter(uploaded))
data = np.frombuffer(uploaded[filename], np.uint8)
img = cv2.imdecode(data, cv2.IMREAD_COLOR)

if img is None:
    raise ValueError("Please upload a valid image file.")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully!")
print("Image size:", gray.shape)
```

Choose files nature.jpg

nature.jpg(image/jpeg) - 27940 bytes, last modified: 13/10/2026 - 100% done

```
Saving nature.jpg to nature.jpg
Image loaded successfully!
Image size: (460, 612)
```

```
plt.figure(figsize=(7, 5))
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```




```
gaussian = cv2.GaussianBlur(gray, (5, 5), 0)

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
```

```
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

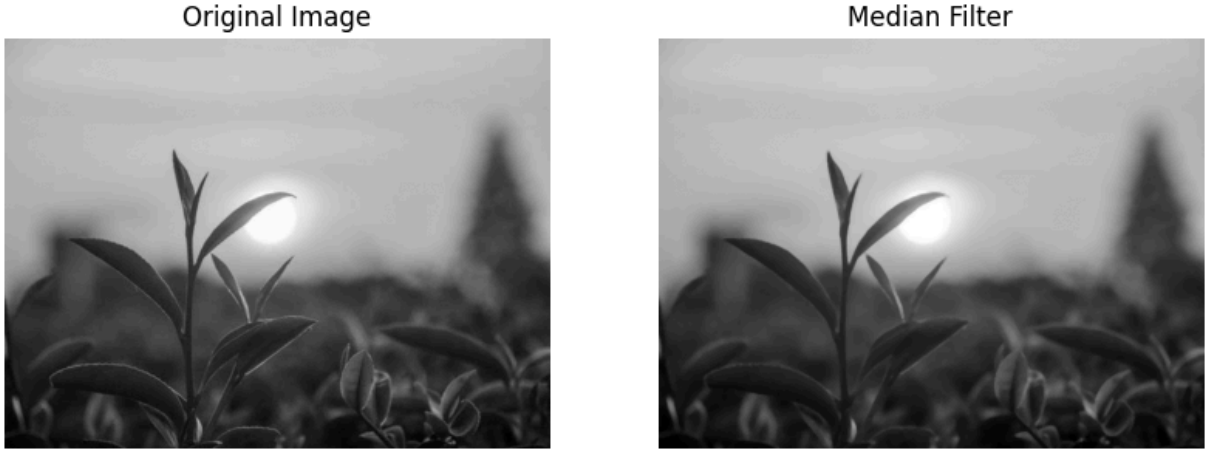
plt.subplot(1, 2, 2)
plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian Blur")
plt.axis("off")
plt.show()
```




```
median = cv2.medianBlur(gray, 5)

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(median, cmap="gray")
plt.title("Median Filter")
plt.axis("off")
plt.show()
```




```
average = cv2.blur(gray, (5, 5))

plt.figure(figsize=(10, 4))
```

```
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

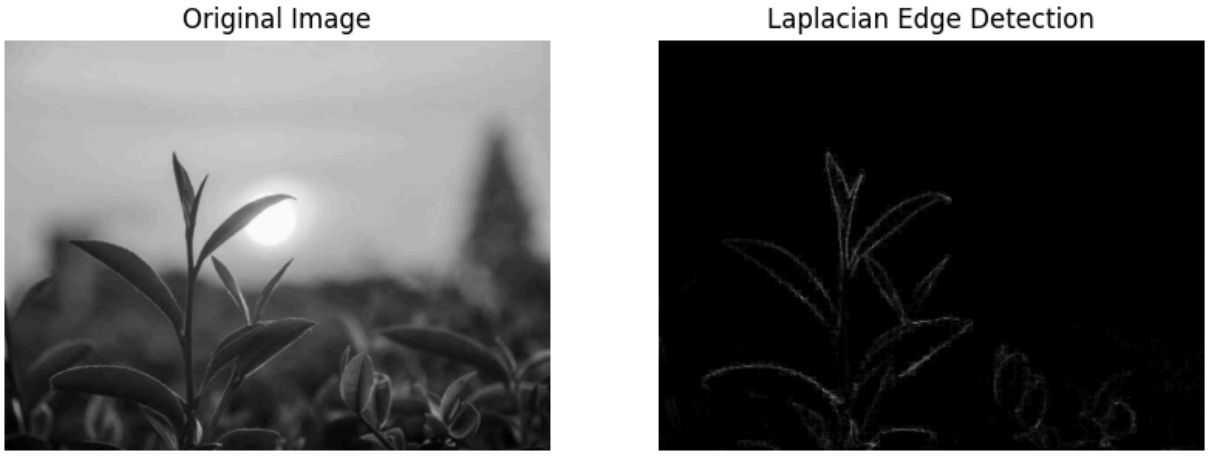
plt.subplot(1, 2, 2)
plt.imshow(average, cmap="gray")
plt.title("Average Filter")
plt.axis("off")
plt.show()
```




```
laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian_display = cv2.convertScaleAbs(laplacian)

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(laplacian_display, cmap="gray")
plt.title("Laplacian Edge Detection")
plt.axis("off")
plt.show()
```




```
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
```

```
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_x_display = cv2.convertScaleAbs(sobel_x)
sobel_y_display = cv2.convertScaleAbs(sobel_y)

sobel_combined = cv2.magnitude(sobel_x, sobel_y)
sobel_combined = cv2.normalize(
    sobel_combined, None, 0, 255, cv2.NORM_MINMAX
).astype(np.uint8)

plt.figure(figsize=(14, 4))

plt.subplot(1, 3, 1)
plt.imshow(sobel_x_display, cmap="gray")
plt.title("Sobel X - Vertical Edges")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(sobel_y_display, cmap="gray")
plt.title("Sobel Y - Horizontal Edges")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(sobel_combined, cmap="gray")
plt.title("Combined Sobel")
plt.axis("off")

plt.tight_layout()
plt.show()
```

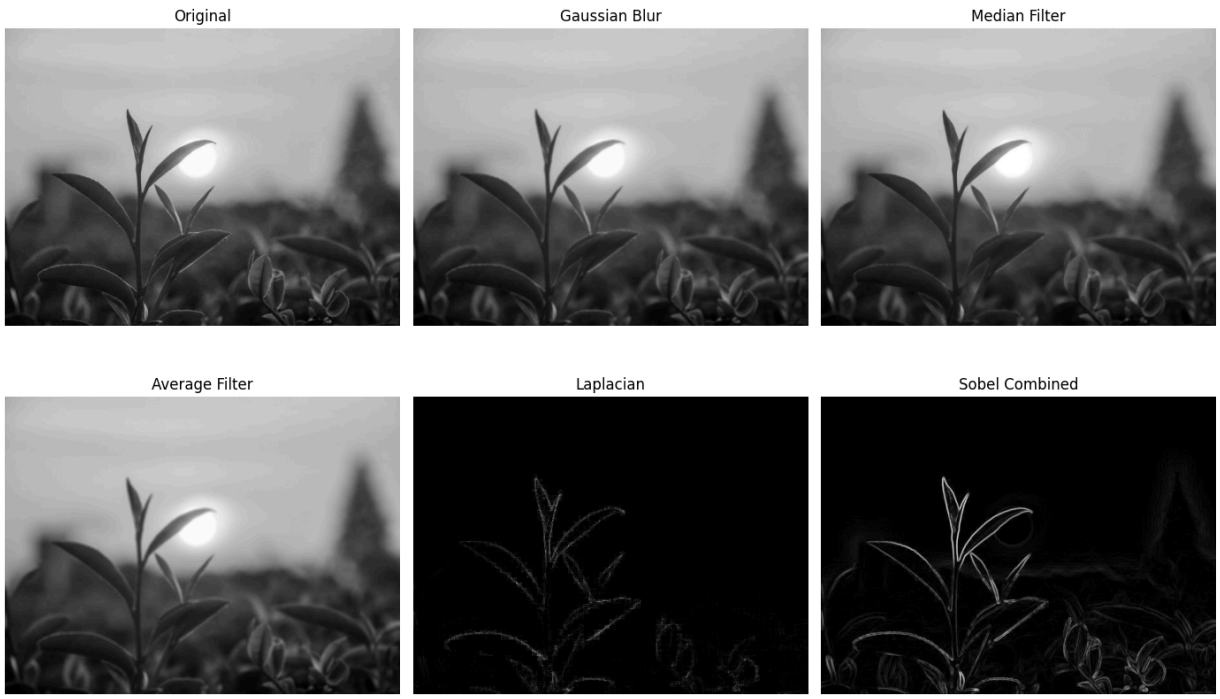



```
results = [
    ("Original", gray),
    ("Gaussian Blur", gaussian),
    ("Median Filter", median),
    ("Average Filter", average),
    ("Laplacian", laplacian_display),
    ("Sobel Combined", sobel_combined)
]

plt.figure(figsize=(14, 9))

for i, (name, result) in enumerate(results, 1):
    plt.subplot(2, 3, i)
    plt.imshow(result, cmap="gray")
    plt.title(name)
    plt.axis("off")

plt.tight_layout()
plt.show()
```




```
print(f"{'Technique':<20} {'Mean Intensity':>16} {'Std Deviation':>16}")
print("-" * 55)

for name, result in results:
    print(
        f"{name:<20} "
        f"{np.mean(result):>16.2f} "
        f"{np.std(result):>16.2f}"
    )

print("\nAnalysis completed successfully.")
```

```
Technique              Mean Intensity    Std Deviation
-------------------------------------------------------
Original                       128.53            66.39
Gaussian Blur                  128.53            66.18
Median Filter                  128.34            66.43
Average Filter                 128.53            66.06
Laplacian                        3.15             8.23
Sobel Combined                   8.53            18.74

Analysis completed successfully.
```

```
qa = [
("Q1. What is spatial filtering? How is it used in digital image processing?",
"Spatial filtering modifies pixel values using neighboring pixels. It is used for smoothing, n

("Q2. Differentiate between Low-Pass Filters and High-Pass Filters with examples.",
"Low-pass filters smooth images and reduce high-frequency noise. Examples include Average, Gau

("Q3. Compare Average Filter, Gaussian Filter, and Median Filter.",
"Average Filter replaces each pixel with the neighborhood average. Gaussian Filter uses a weig

("Q4. Why is the Median Filter effective for removing salt-and-pepper noise?",
"The Median Filter removes isolated black and white pixels by replacing them with the median o

("Q5. Explain the role of convolution kernels in spatial filtering.",
"A convolution kernel is a small matrix of values moved across an image. Its values determine
```

```
("Q6. What is the purpose of Sobel and Laplacian operators in edge detection?",
"The Sobel operator estimates intensity gradients in horizontal and vertical directions. The La

("Q7. Why are filtering operations essential preprocessing steps in computer vision?",
"Filtering can reduce noise, smooth unwanted variations, and highlight important structures. Th

("Q8. Discuss the trade-off between image smoothing and edge preservation.",
"Strong smoothing reduces noise but can blur edges and remove fine details. Weak smoothing pres

("Q9. Mention four real-world applications of spatial filtering.",
"1. Medical image enhancement.\n2. Satellite and remote sensing images.\n3. Surveillance and s

("Q10. Compare spatial domain filtering with frequency domain filtering.",
"Spatial domain filtering directly modifies image pixels using neighborhoods or kernels. Frequ
]

print("EXPERIMENT NO. 3")
print("SPATIAL FILTERING TECHNIQUES")
print("=" * 70)

for question, answer in qa:
    print("\n" + question)
    print("-" * 70)
    print(answer)
    print()

print("All 10 questions answered successfully.")
```

```
EXPERIMENT NO. 3
SPATIAL FILTERING TECHNIQUES
======================================================================

Q1. What is spatial filtering? How is it used in digital image processing?
----------------------------------------------------------------------
Spatial filtering modifies pixel values using neighboring pixels. It is used for smoothing, r


Q2. Differentiate between Low-Pass Filters and High-Pass Filters with examples.
----------------------------------------------------------------------
Low-pass filters smooth images and reduce high-frequency noise. Examples include Average, Gau


Q3. Compare Average Filter, Gaussian Filter, and Median Filter.
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Q4. Why is the Median Filter effective for removing salt-and-pepper noise?
----------------------------------------------------------------------
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Q5. Explain the role of convolution kernels in spatial filtering.
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A convolution kernel is a small matrix of values moved across an image. Its values determine


Q6. What is the purpose of Sobel and Laplacian operators in edge detection?
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Q7. Why are filtering operations essential preprocessing steps in computer vision?
----------------------------------------------------------------------
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Q8. Discuss the trade-off between image smoothing and edge preservation.
----------------------------------------------------------------------
```

```
Strong smoothing reduces noise but can blur edges and remove fine details. Weak smoothing pr


Q9. Mention four real-world applications of spatial filtering.
--------------------------------------------------------------------
1. Medical image enhancement.
2. Satellite and remote sensing images.
3. Surveillance and security systems.
4. Autonomous vehicles and object detection.


Q10. Compare spatial domain filtering with frequency domain filtering.
--------------------------------------------------------------------
Spatial domain filtering directly modifies image pixels using neighborhoods or kernels. Freq

All 10 questions answered successfully.
```